Presenter notes

Slide 1

Timing: 0:00 to 2:30. Establish the business objective, implemented solution and methodology before moving into the technical design.

Branch employees need efficient access to information in operational procedures. The information task includes finding the applicable requirement, understanding its conditions and exceptions, and verifying that the answer is supported by the source. The business objective is to reduce search effort while retaining the detail needed for a correct decision. Productivity improvement is an outcome to measure, not a saving already established by this work.

Branch Assist is a working generative conversational solution over the eleven provided Cayman Bank procedures. The collection contains 139 HTML tables and is organized into 1,388 traceable passages. Questions execute an actual local language model. The interface supports generated answers, contextual follow-ups, procedure selection, citations, source inspection, feedback and conversation export.

The methodology is retrieval-augmented generation. The system first finds relevant procedure evidence, then asks a model to synthesize a concise answer from that evidence. It checks source references, numeric values, recognizable thresholds and claim support before releasing a draft. The source text and check record remain available to the employee. If support is insufficient, the system asks for context or withholds the draft and exposes the original passages.

Four design principles define the solution. Keep each requirement attached to its conditions during document preparation. Retrieve evidence for the current question instead of using a previous generated answer as proof. Make the evidence and response decision inspectable. Evaluate retrieval and response quality separately, including the coverage lost when answers are withheld.

The knowledge boundary is the eleven provided files; external linked procedures and live policy updates are not ingested. The system answers information questions and does not execute bank transactions. The local runtime was selected to work on the available CPU without an API key; its latency and model limitations are measured explicitly.

Success must be judged across accuracy, relevance, completeness and conciseness. Those dimensions drive the evaluation approach later in the presentation. The implemented controls and recorded diagnostics support review; they do not replace independent assessment of substantive instructions.

Transition: the first engineering step is turning the original documents into structured, traceable evidence that retrieval can use.

Slide 2

Timing: 2:30 to 6:00. Explain what happens to all eleven documents and why each preparation decision matters.

The input is a collection of eleven HTML exports with approximately seventy-nine thousand extracted words and 139 tables, including nested tables. Document sizes are uneven, with the business-deposit-account procedure substantially larger than the others. The ingestion process profiles file identity, document title, publication text, section structure and tables before building the searchable collection.

Extraction uses the meaningful policy classes in the HTML exports, including headings that are not ordinary heading tags. It explicitly decodes UTF-8 and retains paragraphs and lists. Scripts and display-only material are excluded, and linked content is not fetched. The original files are preserved so every derived passage can be

checked against its source.

Tables need more than plain text extraction. The parser retains column labels, carries conditions through rowspans, accounts for colspans and includes nested table text once. Oversized row continuations repeat their leftmost condition. This is intended to reduce the risk of separating an action from the situation in which it applies. Text-preservation checks are useful, but do not establish that every table branch has been interpreted correctly.

Narrative passages target roughly 110 whitespace words within a section. Long blocks use a 20-word overlap to reduce context loss at a boundary. Unsplit table rows can reach 130 words, and repeating a condition can make a continuation longer. These are word targets; a separate model-token budget limits the prompt. Shorter passages were selected after larger groupings introduced competing instructions during development. The choice is empirical rather than a universal optimum.

Each passage records its procedure ID, section, source filename, publication text, original-file hash, stable passage identity and table location where relevant. An identical rebuild preserves passage IDs. Source changes require reindexing. The vector matrix is bound to the passage store and checked for shape, finite values, normalization and its own file hash.

The resulting collection has 1,388 passages with a BM25 keyword index and normalized 384-dimensional MiniLM vectors. These complementary indexes support exact terminology and paraphrases. At question time, rankings are combined and reranked before selecting the evidence pack. Retrieval scores rank source matches; they do not estimate the probability that a generated answer is correct.

Transition: the prepared collection now feeds the request architecture, where evidence selection, generation and output checks operate together.

Slide 3

Timing: 6:00–9:30. Explain the request in order, then defend the choices and their limits.

A request first passes basic length, identifier and direct-instruction checks. The resolver identifies an explicit procedure, a new named topic or a continuation of the user's situation. Search scope is enforced before retrieval. Sources and cached index artifacts are checked for changes, including the vector-matrix hash, dimensions and normalization.

BM25 and normalized MiniLM embeddings search the same 1,388 passages. Reciprocal rank fusion combines their top candidates without comparing incompatible raw scores. The ms-marco MiniLM cross-encoder reranks the fused candidates. Weak retrieval signals lead to clarification or insufficient support. The small collection fits exact in-memory vector search, so a managed vector database would add infrastructure without demonstrated benefit here.

Relevant source passages, including a bounded adjacent qualification when applicable, become the evidence pack. Qwen3-4B in Q4_K_M GGUF form generates locally through llama.cpp on CPU. Non-thinking operation, greedy decoding, a 3,800-input-token limit and a 240-output-token limit suit this constrained answer task. The prompt asks for the exact fact first, relevant conditions and citations, using only evidence. Greedy decoding does not guarantee correctness.

Qwen3 was selected after development comparisons with a smaller Qwen2.5 baseline that produced more distracting detail and withheld drafts. This was an end-to-end engineering iteration, not an isolated causal experiment. Local inference avoids API-key and hosted inference dependencies for the local runtime, while adding CPU latency. A stronger approved deployment model still needs comparison before a pilot.

Checks bind claims to their actual citations, compare numeric values and recognizable thresholds, and screen support. Substantive exact wording is matched to the cited passage; paraphrases use a DeBERTa NLI model. Its thresholds are development settings, not calibrated confidence. A failed draft gets at most one correction with the same evidence and failed-check feedback. If it still fails, the draft is withheld and source text remains

available. The screen can also reject a correct answer or miss an error.

This is one observable orchestrator with read-only evidence access, not an autonomous bank transaction agent. Activity records show real actions, counts and timings, not private model reasoning. RAG was chosen over fine-tuning for source replacement and traceability; sending every document would waste context and mix unrelated conditions.

Transition: these controls matter only if an employee can understand and use them in conversation.

Slide 4

Timing: 9:30 to 13:00. Explain the employee workflow and context rules, then use the short recorded dialogue as supporting evidence.

The experience begins with focused access to the procedure collection. Employees can ask directly, use a guided starter, search the procedure library or restrict retrieval to a selected document. Scope is enforced: a question naming a different procedure asks the user to change the selected scope rather than answering from the wrong collection.

Conversation state uses prior user questions to resolve the current situation. Previous generated claims are never factual evidence. A newly named task in the same procedure retains procedure scope while avoiding irrelevant terms from the earlier task. A hypothetical changing a condition keeps the earlier situation, even when the user repeats the procedure name or ID. An explicitly different topic replaces that context and searches the new procedure. Ambiguous or long conversations remain a limitation of the deterministic resolver.

The panel reproduces actual recorded responses. The first question asks about overnight coin storage and receives the $5,250 answer with S1. The follow-up asks about single or dual control for monthly testing. The system retrieves the testing instruction and answers single control with S2. The example illustrates the context rule; the product capability is fresh evidence retrieval for each turn.

Citation buttons select the source passage belonging to that answer. The Evidence desk exposes the procedure, section, publication text, source-file check and original document. Checks shows citation attribution, numeric consistency, recognized thresholds and support screening. Activity shows observed execution stages and elapsed times, with a deterministic explanation of the response decision rather than simulated private model reasoning.

A failed draft may receive one correction using the same evidence and check feedback. If it still fails, the system withholds the draft and leaves the source passages available for review. Weak evidence or missing context produces an explicit insufficient-support or clarification outcome. These controls can still miss errors or reject a correct answer, so the original procedure remains inspectable.

Browser sessions isolate conversation histories. New investigation resets the situation. Employees can mark a response for review and export the questions, answers, evidence, checks, activity and feedback. The export is opt-in; an enterprise audit and retention service is separate deployment work.

Transition: the interface makes responses inspectable; evaluation must determine their accuracy, relevance, completeness and conciseness independently of presentation quality.

Slide 5

Timing: 13:00–17:00. This is the evidence slide. Separate passage retrieval, answer diagnostics, coverage and independent correctness.

The suite contains 30 ordinary questions covering all eleven procedures, four two-turn conversations and six selected boundary cases. Nineteen ordinary questions are development cases and eleven are challenge cases. Labels were AI-authored from the supplied documents and source-checked; after development inspection, the challenge split is not a blind external holdout. Results describe this regression set, not enterprise accuracy.

For the same questions, keyword search finds a labelled relevant passage in the first five results for 27 of 30 cases, semantic search for 28, and hybrid retrieval plus reranking for all 30. Hybrid mean reciprocal rank at five is 0.9278. Finding one labelled passage does not establish that every material exception was retrieved or that the generated answer is correct.

The automatic expected-fact proxy passes 23 of 30 ordinary cases. It checks phrase groups, so it can miss a correct paraphrase and pass a wrong statement containing expected terms. A source-only fallback counts as a failure of generated-answer coverage even if the excerpts contain the answer. Across 35 generated evaluations, six drafts were withheld after checks. The four original final-turn follow-up proxies include two misses; those misses remain in the report. Six deliberately chosen boundary cases matched their expected statuses, which is not a general security assessment.

The table makes the four requested quality dimensions explicit. Accuracy needs an independent reviewer to assess every instruction against the source and stated situation; citations, numeric consistency and NLI are only diagnostics. Relevance asks whether the exact question is answered without unrelated workflow detail. Completeness asks whether applicable conditions, exceptions, warnings and escalation requirements are present. Conciseness asks whether wording is brief without becoming incomplete. The measured mean draft length is 28 words across 32 drafts with recorded length, including withheld drafts; it is not a measure of answer usefulness.

Serial local CPU latency is p50 13.81 and p95 23.92 seconds, excluding initial model load. The engineering suite has 49 passing tests for mechanics and safeguards, and real browser workflows were checked. Neither test count nor latency proves substantive correctness.

Next, two operational subject-matter reviewers should independently score all four dimensions on unseen scenarios, adjudicate disagreements, and report critical errors separately from average scores. Include false refusals, answer coverage, source-version conflicts and long conversations. Compare employee task success and time to correct guidance against existing search before claiming business value.

Transition: these findings shape both what we will demonstrate now and what leadership should validate next.

Slide 6

Timing: 17:00 to 20:00, followed by the separate 15-minute solution walkthrough. Demonstrate the implemented behavior and connect it to the engineering decisions.

The demonstration should show three aspects of the solution: source grounding, conversation continuity and a clear outcome when evidence is insufficient. Use the live application or identify the approved silent recording as a recording. The saved walkthrough lasts 9 minutes 25 seconds and leaves time within the 15-minute segment for live source inspection and technical questions.

Begin with the coin-machine overnight limit and a follow-up about monthly testing. Open the source for each answer and show that the second turn uses the testing passage. Next, ask about notarizing an emailed document and change the condition by making the requester physically present. The supplied restriction remains applicable; the changed condition must not become unsupported permission.

Then ask the savings-bond limit for an account open less than twelve months and follow up about The Private Bank Experience. PR1949 states a $1,000 daily aggregate limit and an exception for eligible customers regardless of account age. Other applicable conditions remain. In the approved recording, the exception draft still fails a check after one correction. The tool therefore exposes the original text for review instead of presenting that draft as an accepted answer.

Also show a topic change, missing context after reset, an unsupported question, review feedback and the evidence export. Use the Evidence, Checks and Activity views to explain how the system reached its release or source-review decision. These are observable actions and screening results; a fluent answer and a passing screen do not establish independent correctness.

The technical conclusions follow from the evidence. Structural ingestion and source identity support traceability. Hybrid retrieval combines exact terminology and paraphrases. A bounded local model call keeps the evidence boundary inspectable, with CPU latency as a tradeoff. Context rules improve continuity while keeping prior generated answers out of the evidence. Output screening increases inspectability but can cause unnecessary withholding and still needs independent assessment.

The next engineering work is independent review of all four response-quality dimensions, broader condition and exception coverage, and stronger-model, latency and concurrency comparisons. Any operational deployment also requires authoritative source versions, employee access controls, monitoring, escalation and rollback. Those capabilities are future work and are not presented as implemented.

Generative AI assisted design, implementation, test-case drafting and presentation preparation, and Qwen powers runtime answering. Validation includes source checks, automated engineering tests, retained response traces and actual browser workflows. AI-authored labels remain subject to independent review.

Handoff: open the prepared tool workflow and ask the first procedure question.